$\boxed{P4.3}$ $\quad Y = \Theta\phi$

$$\phi(\varphi) = e^{im\varphi}, \quad \Theta(\theta) \sim P_\ell^m(\cos\theta)$$

$\ell = m = 0$

$$Y_0^0(\theta,\varphi) = \sqrt{\frac{1}{4\pi}}\, P_0^0(\cos\theta)$$

$$P_0^0(x) = P_0(x), \quad P_0(x) = 1 \quad \Rightarrow \quad Y_0^0 = \frac{1}{\sqrt{4\pi}}.$$

Normalization: $\displaystyle\int_0^\pi \frac{1}{4\pi} \sin\theta\, d\theta \int_0^{2\pi} d\varphi = 1.$

$\ell = 2$
$m = 1$

$$Y_2^1 = (-1)^1 \sqrt{\frac{2\cdot 2+1}{4\pi}\,\frac{(2-1)!}{(2+1)!}}\; e^{i\varphi} P_2^1(\cos\theta)$$

$$P_2^1(x) = (1-x^2)^{1/2}\left(\frac{d}{dx}\right) P_2(x)$$

$$P_2(x) = \frac{1}{2^2 2!}\left(\frac{d}{dx}\right)^2 (x^2-1)^2 = \frac{1}{8}\frac{d^2}{dx^2}(x^4 - 2x^2 + 1) =$$

$$= \frac{1}{8}\left(4\cdot 3\, x^2 - 2\cdot 2\right) = \frac{1}{2}(3x^2 - 1)$$

$$\Rightarrow P_2^1(x) = (1-x^2)^{1/2}\frac{d}{dx}\,\frac{1}{2}(3x^2-1) = 3x\,(1-x^2)^{1/2}$$

$$\Rightarrow P_2^1(\cos\theta) = 3\cos\theta\sqrt{1-\cos^2\theta} = 3\cos\theta\sin\theta$$

$$\Rightarrow Y_2^1(\theta,\varphi) = (-1)\sqrt{\frac{5}{4\pi}\cdot\frac{1}{6}}\; 3e^{i\varphi}\cos\theta\sin\theta$$

$$= -\sqrt{\frac{15}{8\pi}}\, e^{i\varphi}\sin\theta\cos\theta$$

Normalization:

$$\int |Y_2^1|^2 \sin\theta\, d\theta\, d\varphi = 2\pi\left(\frac{15}{8\pi}\right)\int_0^\pi \sin^2\theta\cos^2\theta\,\sin\theta\, d\theta$$

$$= \frac{15}{4}\int_1^{-1}(-)(1-\cos^2\theta)\cos^2\theta\, d(\cos\theta) = \frac{15}{4}\int_{-1}^{1}(1-x^2)\,x^2\,dx =$$

$-\sin\theta\, d\theta = d(\cos\theta)$ $\qquad X = \cos\theta$

$$\int \left(Y_2^1\right)^2 \sin\theta\, d\theta\, d\varphi = \frac{15}{4}\cdot 2\cdot \int_0^1 \left(x^2 - x^4\right) dx =$$

$$= \frac{15}{2}\left(\frac{1}{3}x^3 - \frac{1}{5}x^5\right)\Big|_0^1 = \frac{15}{2}\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{15}{2}\,\frac{5-3}{15} = 1.$$

<u>Orthogonality</u> $\int \left(Y_0^0\right)^* Y_2^1 \sin\theta\, d\theta\, d\varphi =$

$$= \frac{1}{\sqrt{4\pi}}\left(-\sqrt{\frac{15}{8\pi}}\right)\int_0^{\pi} \sin\theta\cos\theta\sin\theta\, d\theta \underbrace{\int_0^{2\pi} e^{i\varphi}\, d\varphi}_{(-i)\, e^{i\varphi}\Big|_0^{2\pi} = 0} = 0$$

$\boxed{P4.7}$ Eq. 4.46: $n_\ell(x) = -(-x)^\ell \left(\frac{1}{x}\frac{d}{dx}\right)^\ell \frac{\cos x}{x}$

$$n_1(x) = -(-x)\,\frac{1}{x}\frac{d}{dx}\,\frac{\cos x}{x} = \frac{d}{dx}\,\frac{\cos x}{x} =$$

$$= -\frac{\sin x}{x} - \frac{\cos x}{x^2}$$

$$n_2(x) = -x^2\left(\frac{1}{x}\frac{d}{dx}\right)\left(\frac{1}{x}\frac{d}{dx}\right)\frac{\cos x}{x} =$$

$$= -x\,\frac{d}{dx}\,\frac{1}{x}\left(-\frac{\sin x}{x} - \frac{\cos x}{x^2}\right) = x\,\frac{d}{dx}\left(\frac{\cos x}{x^3} + \frac{\sin x}{x^2}\right) =$$

$$= x\left(-\frac{\sin x}{x^3} - \frac{3\cos x}{x^4} + \frac{\cos x}{x^2} - \frac{2\sin x}{x^3}\right) =$$

$$= -\left(\frac{3\sin x}{x^2} + \frac{3\cos x}{x^3} - \frac{\cos x}{x}\right).$$

(b) $n_1(x) \underset{x\to 0}{\longrightarrow} \frac{-x}{x} - \frac{1}{x^2} = -\frac{1}{x^2} \underset{x\to 0}{\longrightarrow} \infty$

$$n_2(x) \underset{x\to 0}{\longrightarrow} (-)\left(\frac{3x}{x^2} + \frac{3}{x^3} - \frac{1}{x}\right) = \frac{-3}{x^3} \underset{x\to 0}{\longrightarrow} \infty$$

$\boxed{4.8}$

(a) Radial Equation :

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left[V + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right]u = Eu$$

$$u = r\, j_1(kr) = r\left[\frac{\sin k_{\hbar} r}{(k_{\hbar} r)^2} - \frac{\cos kr}{kr}\right] = \frac{1}{k}\left[\frac{\sin kr}{kr} - \cos kr\right]$$

$l=1$, $V=0$ and put $kr = r' \Rightarrow u = \frac{1}{k}\left[\frac{\sin r'}{r'} - \cos r'\right]$

thus eqn modifies to →

$$-\frac{\hbar^2}{2m}k^2\frac{d^2u}{dr'^2} + \left[\frac{\hbar^2}{2m}k^2\frac{(2)}{r'^2}\right]u = Eu$$

$$\frac{du}{dr'} = \frac{1}{k}\left[\frac{\cos r'}{r'} - \frac{\sin r'}{r'^2} + \sin r'\right]$$

$$\frac{d^2u}{dr'^2} = \frac{1}{k}\left[-\frac{\cos r'}{r'^2} - \frac{\sin r'}{r'} - \frac{\cos r'}{r'^2} + 2\frac{\sin r'}{r'^3} + \cos r'\right]$$

$$= \frac{1}{k}\left[2\frac{\sin r'}{r'^3} - 2\frac{\cos r'}{r'^2}\right] - u = \frac{2}{r'^2}u - u$$

$$\Rightarrow -\frac{\hbar^2}{2m}k^2\left[\frac{2u}{r'^2} - u\right] + \frac{\hbar^2}{2m}\frac{k^2(2)}{r'^2}u = \frac{\hbar^2k^2}{2m}u = Eu$$

$$\Rightarrow E = \frac{\hbar^2k^2}{2m}$$

(b) For infinite spherical well boundary conditions

(4)

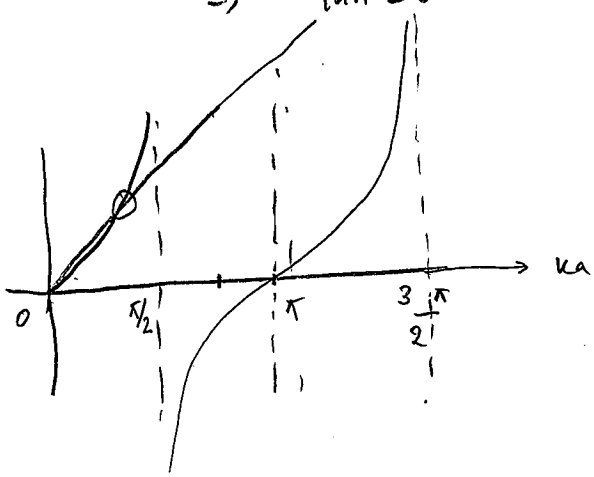
determines the allowed K-values and consequently the Energies →

the Boundary Condition →

$$U(r=a) = a j_1(ka) = 0$$

$$\Rightarrow \frac{1}{k}\left[\frac{\sin ka}{ka} - \cos ka\right] = 0$$

$$\Rightarrow \tan ka = ka$$



Clearly for $k_n a \gg 1$ or large n

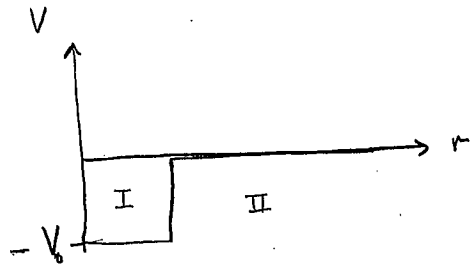
intersection takes place at

$$k_n a \approx (2n+1)\pi/2 \Rightarrow k_n \approx \frac{(2n+1)\pi}{2a}$$

$$\Rightarrow \varepsilon = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 \pi^2}{2ma^2}\left(n+\frac{1}{2}\right)^2.$$

$\boxed{4.9}$

$$V(r) = \begin{cases} -V_0 & \text{if } r \le a \\ 0 & \text{if } r > a \end{cases}$$



$\boxed{l = 0}$

Looking for Bound State : $E = -|E|$

$\Rightarrow$ In region I : $V = -V_0$

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left[-V_0\right]u = -|E|\,u$$

$$u_I = A\sin kx + B\sin kx \;;\quad \text{at } r=0 \quad u(r=0)=0$$

$$\Rightarrow \quad B = 0 \quad \text{and} \quad \frac{\hbar^2k^2}{2m} - V_0 = -|E| \Rightarrow \frac{\hbar^2k^2}{2m} = V_0 - |E|$$

$\Rightarrow$ In region II : $V = 0$

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} = -|E|\,u$$

$$\Rightarrow \quad u_{II} = c\,e^{-lx} \;;\quad \frac{\hbar^2 l^2}{2m} = |E| = V_0 - \frac{\hbar^2k^2}{2m}$$

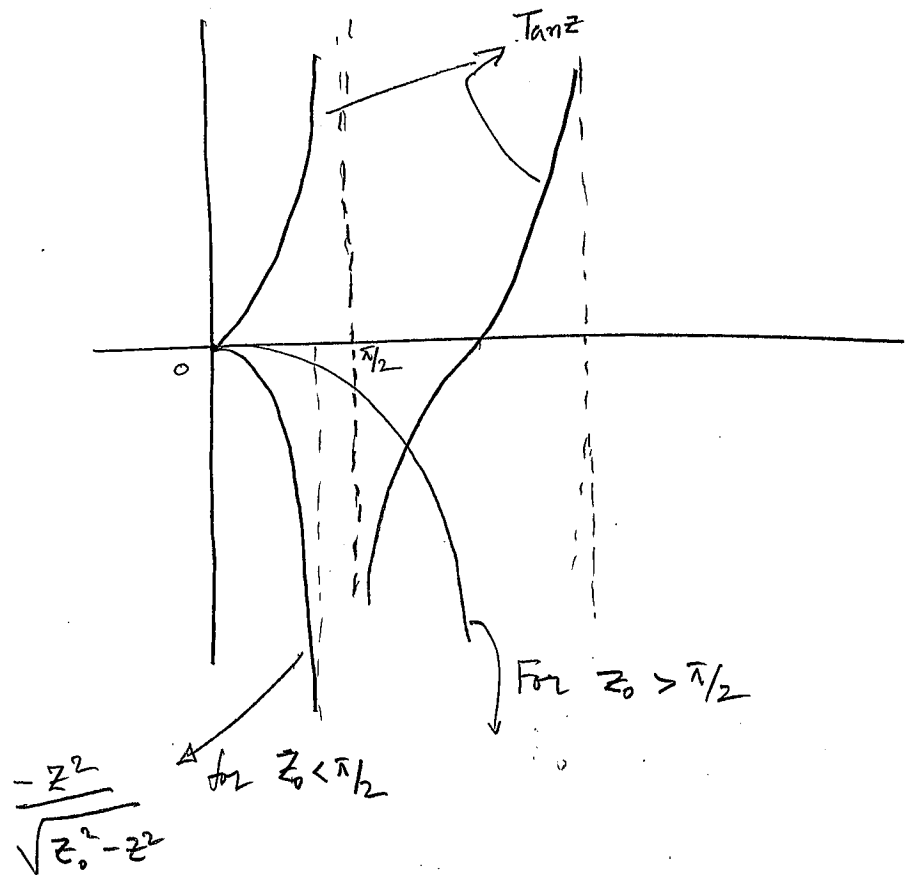
at $r=a \rightarrow$ $u_I(a) = u_{II}(a)$ and $\left.\frac{d}{dr}u_I(x)\right|_{r=a_-} = \left.\frac{d}{dr}u_{II}(x)\right|_{r=a_+}$

$$\Rightarrow A\sin ka = c\,e^{-la} \qquad A\,k\cos ka = -c\,l\,e^{-la}$$

$$\tan ka = -\frac{k}{l} = -\frac{1}{\sqrt{\frac{2mV_0a^2}{\hbar^2 k^2a^2} - 1}}$$

put $ka = z$

$$\Rightarrow \quad \tan z = -\frac{z^2}{\sqrt{z_0^2 - z^2}} \; ; \quad \text{where } z_0^2 = \frac{2mV_0a^2}{\hbar^2}$$



Clearly $\tan z = -\frac{z^2}{\sqrt{z_0^2 - z^2}}$ has no solutions if $z_0 < \pi/2$

i.e if $V_0 a^2 < \frac{\pi^2 \hbar^2}{8m}$.